Version 7.4.7

PVsyst - Simulation report

Grid-Connected System

Project: RooftopCI_Hooghly_307kWp

Variant:

No 3D scene defined, no shadings

System power: 307 kWp

Kolkata - India

Author

PVsyst V7.4.7
VC0, Simulation date:
05/11/26 12:45
with V7.4.7

Project summary

Geographical Site
Kolkata
India

Situation

Latitude	22.56 °N
Longitude	88.36 °E
Altitude	11 m
Time zone	UTC+5.5

Project settings

Albedo	0.20

Weather data
Kolkata
Meteonorm 8.1 (1991-2012) - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation

Fixed planes	2 orientations
Tilts/azimuths	7 / -90 ° 7 / 90 °

Near Shadings
No Shadings

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	544 units
Pnom total	307 kWp

Inverters

Nb. of units	2 units
Pnom total	220 kWac
Pnom ratio	1.397

Results summary

Produced Energy	409153 kWh/year	Specific production	1331 kWh/kWp/year	Perf. Ratio PR	84.90 %

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

PV Field Orientation

Orientation		Sheds configuration	Models used	
Fixed planes	2 orientations	No 3D scene defined	Transposition	Perez
Tilts/azimuths	7 / -90 ° 7 / 90 °		Diffuse	Perez, Meteonorm
			Circumsolar	separate

Horizon	Near Shadings	User's needs
Free Horizon	No Shadings	Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Waaree Energies
Model	BiN-08-565
(Custom parameters definition)	
Unit Nom. Power	565 Wp
Number of PV modules	544 units
Nominal (STC)	307 kWp

Inverter

Manufacturer	Sungrow
Model	SG110CX-P2
(Original PVsyst database)	
Unit Nom. Power	110 kWac
Number of inverters	2 units
Total power	220 kWac

Array #1 - East slope

Orientation	#1		
Tilt/Azimuth	7/-90 °		
Number of PV modules	272 units	Number of inverters	1 unit
Nominal (STC)	154 kWp	Total power	110 kWac
Modules	16 string x 17 In series		
At operating cond. (50°C)		Operating voltage	180-1000 V
Pmpp	142 kWp	Pnom ratio (DC:AC)	1.40
U mpp	676 V	Power sharing within this inverter	
I mpp	210 A		

Array #2 - West slope

Orientation	#2		
Tilt/Azimuth	7/90 °		
Number of PV modules	272 units	Number of inverters	1 unit
Nominal (STC)	154 kWp	Total power	110 kWac
Modules	16 string x 17 In series		
At operating cond. (50°C)		Operating voltage	180-1000 V
Pmpp	142 kWp	Pnom ratio (DC:AC)	1.40
U mpp	676 V	Power sharing within this inverter	
I mpp	210 A		

Total PV power		**Total inverter power**	
Nominal (STC)	307 kWp	Total power	220 kWac
Total	544 modules	Number of inverters	2 units
Module area	1405 m²	Pnom ratio	1.40

Array losses

Thermal Loss factor

Module temperature according to irradiance

Uc (const)	20.0	W/m²K
Uv (wind)	0.0	W/m²K/m/s

DC wiring losses

Global array res.	53	mΩ
Global wiring resistance	26	mΩ
Loss Fraction	1.5	% at STC

Module Quality Loss

Loss Fraction	-0.5 %

Module mismatch losses

Loss Fraction	2.0	% at MPP

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

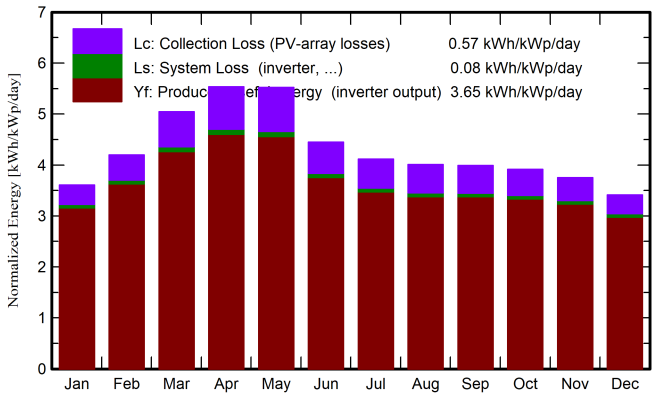
0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000

Main results

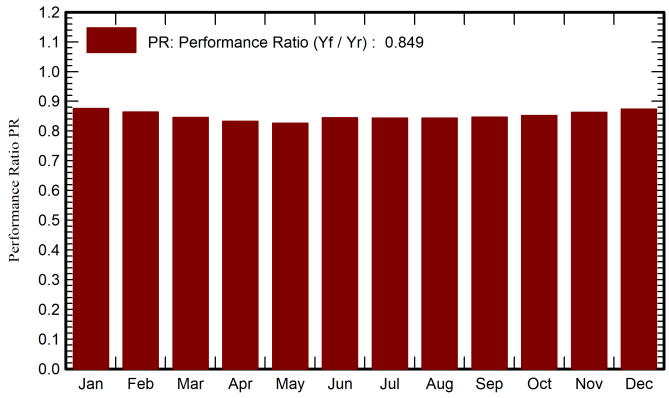
System Production

Produced Energy	409153 kWh/year	Specific production	1331 kWh/kWp/year
		Perf. Ratio PR	84.90 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

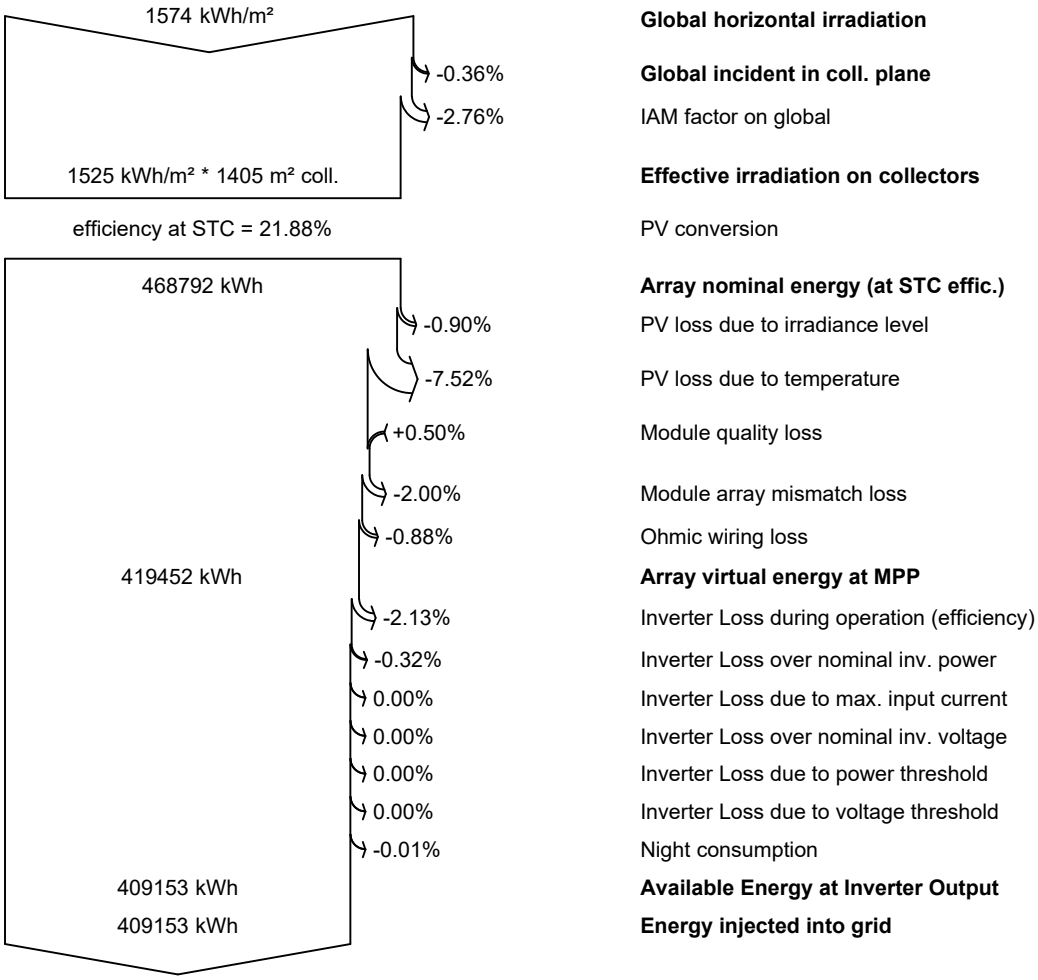
	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	112.1	60.55	17.45	111.8	107.8	30773	30104	0.876
February	118.0	67.00	22.10	117.6	114.2	31940	31248	0.864
March	156.9	85.04	27.55	156.5	152.6	41565	40680	0.846
April	166.5	88.89	30.18	166.0	162.3	43412	42480	0.833
May	171.7	96.70	31.43	171.2	167.4	44439	43496	0.827
June	134.0	93.63	30.47	133.5	129.9	35389	34641	0.844
July	128.3	87.28	29.78	127.7	124.3	33818	33089	0.843
August	125.0	83.58	29.36	124.4	121.1	32940	32226	0.843
September	120.4	78.22	28.31	119.7	116.4	31844	31174	0.847
October	121.8	73.59	27.25	121.4	117.8	32475	31788	0.852
November	112.8	60.86	23.36	112.5	108.8	30477	29823	0.862
December	106.1	57.07	18.68	105.8	102.0	29053	28404	0.874
Year	1573.6	932.42	26.34	1568.0	1524.7	418123	409153	0.849

Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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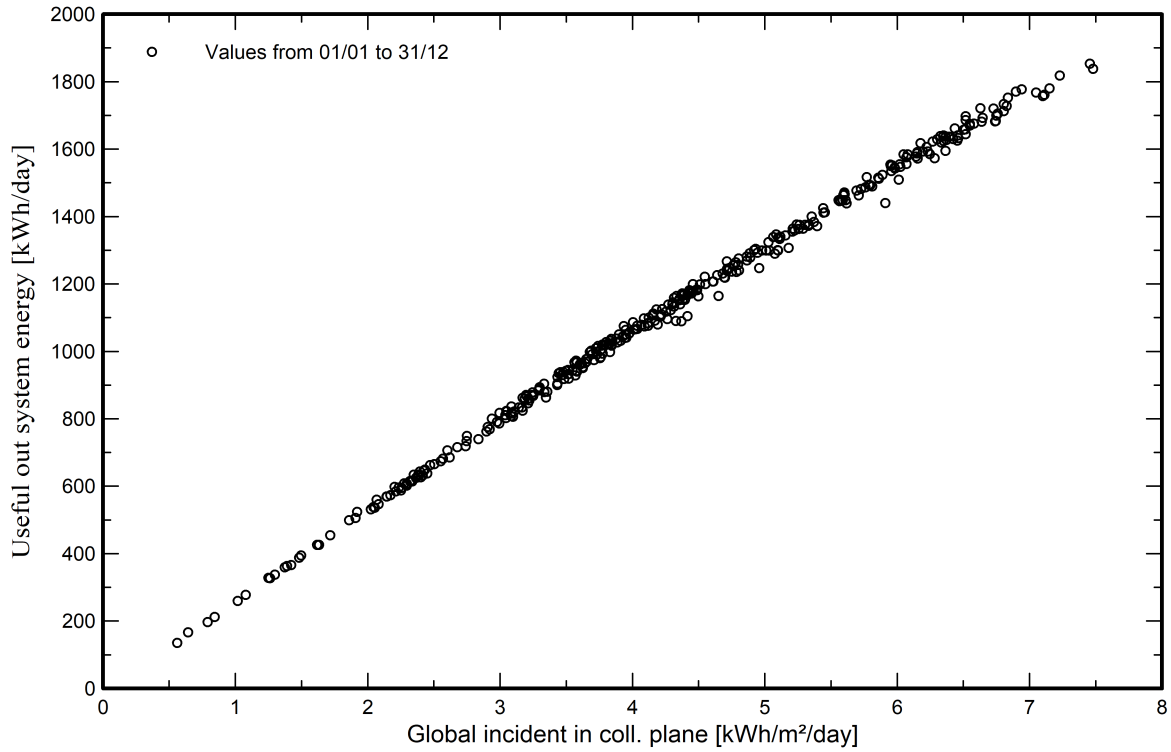
Loss diagram



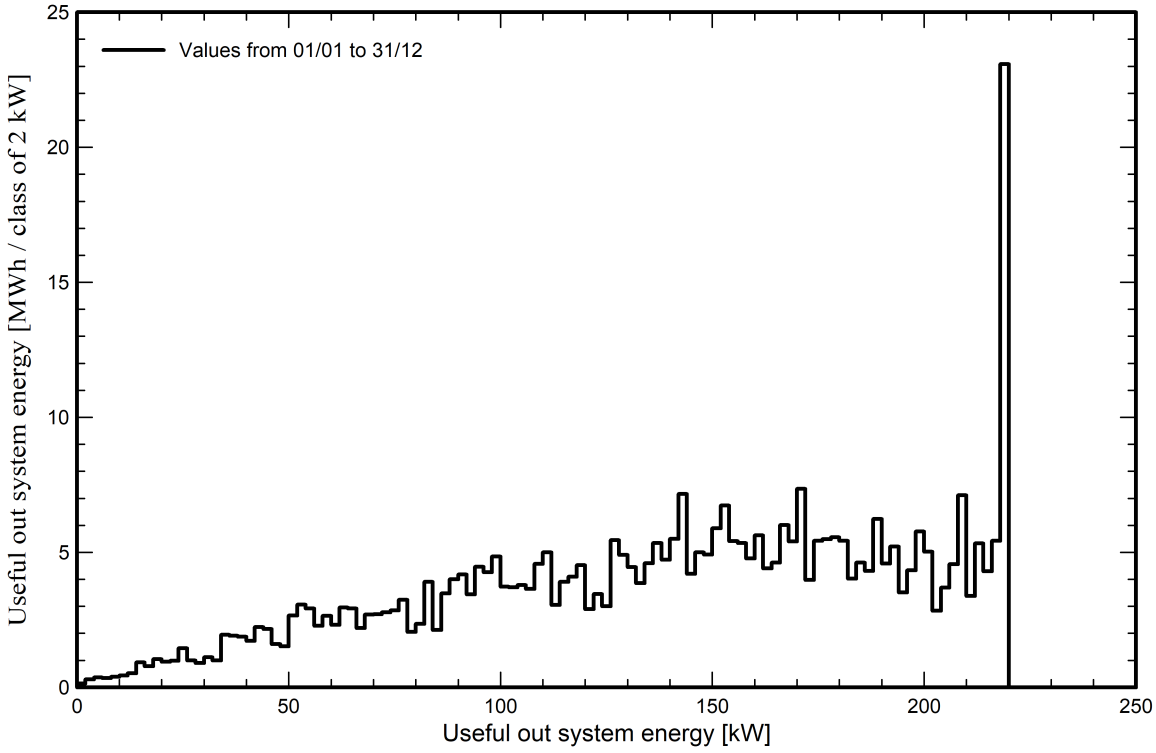
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Predef. graphs

Daily Input/Output diagram

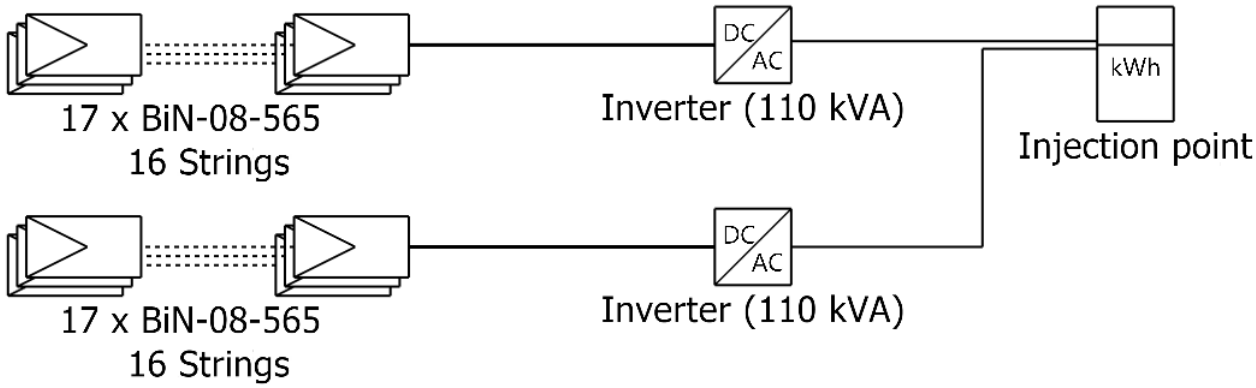


System Output Power Distribution



Single-line diagram

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PV module	BiN-08-565
Inverter	SG110CX-P2
String	17 x BiN-08-565

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